Working Notes for $D_{s1} \to D_s\gamma$ in Light-Cone QCD Sum Rules

Ds gamma project

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Abstract

These notes are a living derivation for the light-cone QCD sum-rule calculation of $D_{s1}(2460) \to D_s\gamma$ and $D_{s1}(2536) \to D_s\gamma$. The purpose is pedagogical as well as practical: each step is written so that a master's student can follow the choices, conventions, and algebra before the final paper-level formulas are assembled.

1 Conventions

We use the mostly-minus metric

$$g_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \tag{1}$$

and

$$\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \qquad \sigma_{\mu\nu} = \frac{\mathrm{i}}{2}[\gamma_\mu, \gamma_\nu], \qquad \gamma_5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3. \tag{2}$$

The Levi-Civita convention is

$$\epsilon^{0123} = +1, \qquad \epsilon_{0123} = -1, \tag{3}$$

so that

$$\mathrm{Tr}[\gamma_5\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta] = -4\mathrm{i}\,\epsilon_{\mu\nu\alpha\beta}. \tag{4}$$

Our Fourier convention for a propagator is

$$S(x) = \int \frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\,\widetilde{S}(k). \tag{5}$$

2 Currents and Momentum Assignment

The external momenta are

$$P = p + q, \tag{6}$$

where P is the initial axial-meson momentum, p is the final pseudoscalar D_s momentum, and q is the real-photon momentum.

The pseudoscalar interpolating current is

$$J_{D_s}(x) = \bar{s}(x)\,\mathrm{i}\gamma_5\,Q(x). \tag{7}$$

For the axial channel we use two independent currents,

$$J^{(0)}_{A\mu}(x) = \bar{s}(x)\gamma_\mu\gamma_5 Q(x), \tag{8}$$

$$J^{(0)}_{B\mu}(x) = \frac{\mathrm{i}}{m_Q + m_s}\,\bar{s}(x)\sigma_{\mu\alpha}P^\alpha\gamma_5 Q(x). \tag{9}$$

The tensor current contracts with P^α, not p^α, because it interpolates the initial axial state. The factor $1/(m_Q + m_s)$ restores the same mass dimension as the axial-vector current.

3 Correlation Function

For a generic axial current $\Gamma_\mu \in \{\gamma_\mu\gamma_5,\, \mathrm{i}\sigma_{\mu\alpha}P^\alpha\gamma_5/(m_Q+m_s)\}$ we study

$$\Pi_\mu(p,q) = \mathrm{i}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\langle\gamma(q,\varepsilon)|T\{\bar{s}(x)\Gamma_\mu Q(x)\,\bar{Q}(0)\mathrm{i}\gamma_5 s(0)\}|0\rangle. \tag{10}$$

The target Lorentz structure for the $1^+\to 0^-\gamma$ transition is

$$(\eta\cdot\varepsilon^*)(p\cdot q) - (\eta\cdot q)(p\cdot\varepsilon^*). \tag{11}$$

On the correlator side we therefore project onto an equivalent transverse structure built from the open axial index and the photon polarization.

4 Wick Contraction and OPE Bookkeeping

The OPE is separated into a hard-photon sector and a soft-photon sector:

$$\Pi_\mu^{\mathrm{OPE}} = \Pi_\mu^{\mathrm{hard}} + \Pi_\mu^{\mathrm{soft}}. \tag{12}$$

This split is important because the two terms describe different physics and must not be double counted.

4.1 Hard photon emission

In the hard sector the photon is emitted perturbatively from either the heavy quark or the strange quark. After Wick contraction one obtains

$$\Pi_\mu^{\mathrm{hard}} = \mathrm{i}N_c\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\mathrm{Tr}_D\Big[\Gamma_\mu S_Q^\gamma(x)\,\mathrm{i}\gamma_5\, S_s^0(-x) + \Gamma_\mu S_Q^0(x)\,\mathrm{i}\gamma_5\, S_s^\gamma(-x)\Big], \tag{13}$$

where $N_c = 3$ and Tr_D denotes the Dirac trace. The free massive propagator is

$$S_q^0(x) = \int \frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\frac{\mathrm{i}(\not{k}+m_q)}{k^2-m_q^2}. \tag{14}$$

We keep explicit powers of m_s, including possible m_s^2 terms.

The first-order electromagnetic background-field correction for a generic quark flavor q is

$$S_q^\gamma(x) = -\mathrm{i}e_q\int\frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\int_0^1 du\left[\frac{\not{k}+m_q}{2(m_q^2-k^2)^2}F^{\alpha\beta}(ux)\sigma_{\alpha\beta} + \frac{ux_\alpha}{m_q^2-k^2}F^{\alpha\beta}(ux)\gamma_\beta\right]. \tag{15}$$

In Eq. (13) this generic formula is used twice:

$$S_Q^\gamma(x) = S_q^\gamma(x)\big|_{q\to Q,\, m_q\to m_Q,\, e_q\to e_Q}, \tag{16}$$

$$S_s^\gamma(x) = S_q^\gamma(x)\big|_{q\to s,\, m_q\to m_s,\, e_q\to e_s}. \tag{17}$$

Thus the hard sector contains both heavy-line and strange-line electromagnetic background-field insertions. For a real photon we write

$$F^{\alpha\beta}(ux) = \mathrm{i}(\varepsilon^\alpha q^\beta - \varepsilon^\beta q^\alpha)\mathrm{e}^{\mathrm{i}uq\cdot x}, \tag{18}$$

with the overall phase kept consistent with the Fourier convention.

Equation (15) is not a vacuum-condensate correction. It is the quark propagator in an external electromagnetic field and is included in the base analysis. What we do not include in the base hard sector are the ordinary SVZ-style condensate-expanded propagator terms such as $\langle\bar{s}s\rangle$, $\langle\bar{s}g_s\sigma Gs\rangle$, or $\langle G^2\rangle$ propagator corrections.

4.2 Equivalent momentum-space vertex form

For FeynCalc trace checks it is often simpler to use the equivalent explicit photon-vertex representation. With photon index ν,

$$N_{Q,\mu\nu} = \mathrm{Tr}_D[\Gamma_\mu(\not{k}+\not{q}+m_Q)\gamma_\nu(\not{k}+m_Q)\mathrm{i}\gamma_5(\not{k}-\not{p}+m_s)]\,, \tag{19}$$

$$N_{s,\mu\nu} = \mathrm{Tr}_D[\Gamma_\mu(\not{k}+m_Q)\mathrm{i}\gamma_5(\not{k}-\not{p}+m_s)\gamma_\nu(\not{k}-\not{p}+\not{q}+m_s)]\,. \tag{20}$$

These numerators are accompanied by the denominators

$$D_Q = [(k+q)^2 - m_Q^2][k^2 - m_Q^2][(k-p)^2 - m_s^2], \tag{21}$$

$$D_s = [k^2 - m_Q^2][(k-p)^2 - m_s^2][(k-p+q)^2 - m_s^2]. \tag{22}$$

The explicit-vertex form and the background-field form represent the same hard emission physics. We must not add both as independent contributions.

4.3 Soft photon emission: two-particle photon DAs

In the soft sector the photon is represented by its light-cone distribution amplitudes. Contracting only the heavy quark gives

$$\Pi_\mu^{\mathrm{soft},2p} = \mathrm{i}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}[\Gamma_\mu S_Q^0(x)\mathrm{i}\gamma_5]_{\alpha\beta}\langle\gamma(q,\varepsilon)|\bar{s}_\alpha(x)s_\beta(0)|0\rangle. \tag{23}$$

Fierz rearrangement converts the open spinor indices to a basis of Dirac bilinears,

$$\bar{s}_\alpha(x)s_\beta(0) = -\frac{1}{4}\sum_i(\Gamma_i)_{\beta\alpha}\bar{s}(x)\Gamma_i s(0), \tag{24}$$

so that

$$\Pi_\mu^{\mathrm{soft},2p} = -\frac{\mathrm{i}}{4}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\sum_i \mathrm{Tr}_D[\Gamma_\mu S_Q^0(x)\mathrm{i}\gamma_5\Gamma_i]\,\langle\gamma(q,\varepsilon)|\bar{s}(x)\Gamma_i s(0)|0\rangle. \tag{25}$$

The photon matrix elements are then expressed in terms of two-particle photon DAs. Parameters such as $\chi\langle\bar{s}s\rangle$ enter here as part of the photon wavefunction normalization; they are not condensate corrections to the hard propagator.

4.4 Soft photon emission: three-particle photon DAs

To include three-particle photon DAs we must also allow one background gluon to come from the heavy-quark propagator,

$$S_Q^G(x) = -\mathrm{i}g_s\int\frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\int_0^1 du\left[\frac{\not{k}+m_Q}{2(m_Q^2-k^2)^2}G^{\alpha\beta}(ux)\sigma_{\alpha\beta} + \frac{ux_\alpha}{m_Q^2-k^2}G^{\alpha\beta}(ux)\gamma_\beta\right]. \tag{26}$$

This term is not a gluon-condensate correction. It is an operator insertion that combines with photon matrix elements of the form

$$\langle\gamma(q,\varepsilon)|\bar{s}(x)\Gamma_i g_s G_{\alpha\beta}(ux)s(0)|0\rangle, \tag{27}$$

which define the three-particle photon DAs.

5 Staged Calculation Plan

We proceed in two stages.

Stage 1. Include the hard photon emission terms and the two-particle photon DAs:

$$\Pi_\mu^{\text{Stage 1}} = \Pi_\mu^{\text{hard}} + \Pi_\mu^{\text{soft},2p}. \tag{28}$$

This stage keeps all explicit m_s terms and omits ordinary condensate-expanded propagators.

Stage 2. Add the three-particle photon DA contributions generated by S_Q^G and the corresponding light-cone matrix elements:

$$\Pi_\mu^{\text{Stage 2}} = \Pi_\mu^{\text{Stage 1}} + \Pi_\mu^{\text{soft},3p}. \tag{29}$$

This staging keeps the derivation auditable: first we verify the core hard and two-particle DA structures, then we add the more involved background-gluon terms.

6 Current FeynCalc Artifacts

The script `scripts/step2_current_building_blocks.wl` checks the basic Dirac conventions and defines the current vertices. The script `scripts/step2_hard_photon_numerators.wl` computes numerator-level hard-photon traces in the explicit photon-vertex representation. These traces are stored in `outputs/step2_hard_photon_numerators.txt`.

The script `scripts/step2_e1_project_hard_numerators.wl` applies the E1 projector

$$\mathcal{P}_{E1}^{\mu\nu} = \frac{p^\nu q^\mu - (p\cdot q) g^{\mu\nu}}{2(p\cdot q)^2}, \tag{30}$$

which satisfies $\mathcal{P}_{E1}^{\mu\nu}[p_\nu q_\mu - (p\cdot q) g_{\mu\nu}] = 1$ for $q^2 = 0$. The projected hard numerators are stored in `outputs/step2_e1_projected_hard_numerators.txt`.

The script `scripts/step2_soft_2p_fierz_traces.wl` computes the trace factors multiplying the two-particle photon DA matrix elements after the Fierz rearrangement. The output `outputs/step2_soft_2p_fierz_traces.txt` shows which bilinears feed the axial current and which feed the tensor current.

7 Stage 1 Axial-Current Numerical Baseline

Before deriving the tensor-current sum rule, we implemented the known single-axial-current result for $D_{s1}(2460) \to D_s\gamma$ as a benchmark. This is the coupling called g_1 by Colangelo, De Fazio, and Ozpineci. In our notation this is the axial-current form factor G_A.

The Stage 1 version contains

$$G_A^{\text{Stage 1}} = G_A^{\text{pert}} + G_A^{\chi\phi_\gamma} + G_A^{\mathbb{A},\bar{H}_\gamma} + G_A^{f_{3\gamma}\Psi^v}, \tag{31}$$

where the three-particle DA integral usually denoted by $\mathcal{F}_1$ is omitted until Stage 2.

The numerical script is

`scripts/stage1_axial_g1_baseline.py`.

It writes the full Borel/threshold grid to

`outputs/stage1_axial_g1_baseline.csv`.

7.1 Magnetic-susceptibility sign convention

The Colangelo paper prints $\chi = -(3.15 \pm 0.30)\,\mathrm{GeV}^{-2}$ in its matrix-element convention. In this project we store the BBK magnitude as

$$\chi(1\,\mathrm{GeV}) = +3.15 \pm 0.30\,\mathrm{GeV}^{-2}. \tag{32}$$

Therefore the sign of the $\chi\phi_\gamma$ term in the implemented sum rule is flipped relative to the literal printed formula. This is a convention translation, not a physical change. A diagnostic run that inserts positive χ into the printed negative-χ formula gives a much larger width and is kept only as a warning check.

7.2 Preliminary Stage 1 result

Using the project inputs, $u_0 = 1/2$, $3 \le M^2 \le 6\,\mathrm{GeV}^2$, and $s_0 = (2.50, 2.55, 2.60)^2\,\mathrm{GeV}^2$, the axial-current Stage 1 grid gives

$$-0.399 \le G_A^{\text{Stage 1}} \le -0.314 \quad \mathrm{GeV}^{-1}. \tag{33}$$

With

$$\Gamma = \frac{\alpha_{\text{em}}}{3} G^2 |\vec{q}_\gamma|^3, \tag{34}$$

this corresponds to

$$20.8\,\mathrm{keV} \le \Gamma_A^{\text{Stage 1}}[D_{s1}(2460) \to D_s\gamma] \le 33.5\,\mathrm{keV}. \tag{35}$$

This is not yet the physical mixed-state result. It is the axial-current baseline that will be combined with the tensor-current form factor after the J_B sum rule is derived.

8 First Tensor-Current Step: Leading-Twist Soft Contribution

The tensor-current calculation starts from the trace-level identity produced by `scripts/step3_soft_2p_e1_pro`
For the two-particle tensor photon DA matrix element, the E1 projections are

$$\mathcal{P}_{E1}\Big[\mathrm{Tr}(J_A\, S_Q\, i\gamma_5\, \sigma_{\rho\lambda})(\varepsilon^\rho q^\lambda - \varepsilon^\lambda q^\rho)\Big] = 8\,\frac{k \cdot q}{p \cdot q}, \tag{36}$$

$$\mathcal{P}_{E1}\Big[\mathrm{Tr}(J_B\, S_Q\, i\gamma_5\, \sigma_{\rho\lambda})(\varepsilon^\rho q^\lambda - \varepsilon^\lambda q^\rho)\Big] = 8\,\frac{m_Q}{m_Q + m_s}. \tag{37}$$

After the two-particle DA Fourier integral and double Borel transform, $k = p + \bar{u}q$, hence $k \cdot q = p \cdot q$ for $q^2 = 0$. Therefore, before the hadronic decay-constant normalization, the leading-twist tensor-current OPE piece is

$$\Pi_B^{\chi\phi_\gamma} = \frac{m_Q}{m_Q + m_s}\,\Pi_A^{\chi\phi_\gamma}. \tag{38}$$

The exploratory script `scripts/stage1_tensor_gb_leading_twist.py` uses this relation to estimate only the $\chi\phi_\gamma$ part of G_B. This is not the final tensor-current sum rule. It omits the tensor-current hard contribution, the vector DA contribution, and the Stage 2 three-particle DA terms.

Using the project inputs, the partial leading-twist estimate gives

$$G_B^{\chi\phi_\gamma} \simeq +0.205 \text{ to } +0.273\ \mathrm{GeV}^{-1}, \qquad \text{if } f_B = f_T, \tag{39}$$

$$G_B^{\chi\phi_\gamma} \simeq +0.114 \text{ to } +0.151\ \mathrm{GeV}^{-1}, \qquad \text{if } f_B = f_T\,\frac{m_{D_{s1}}}{m_c + m_s}. \tag{40}$$

The sizeable sign difference relative to G_A means that the physical mixed amplitudes can be sensitive to the tensor-current normalization. This is why the next step is to finish the full G_B sum rule before quoting final mixed widths.

9 Decay-Constant Normalization Before Mixing

The labels f_A, f_B, and f_T must not be mixed casually. The axial current couples to the axial basis state through

$$\langle 0|J_A^\mu|D_{s1}^A(P,\eta)\rangle = f_A m_A \eta^\mu, \tag{41}$$

whereas our tensor-derived current is

$$J_B^\mu = \frac{\mathrm{i}}{m_c + m_s}\,\bar{s}\sigma^{\mu\alpha}P_\alpha\gamma_5 c. \tag{42}$$

The corresponding decay constant is defined by

$$\langle 0|J_B^\mu|D_{s1}^B(P,\eta)\rangle = f_B m_B \eta^\mu. \tag{43}$$

Therefore $f_B = f_T$ is not a physical statement by itself. It is only true if the tabulated f_T is already defined for the same current J_B^μ and with the same mass normalization. If instead the input table gives the more common tensor-current matrix element

$$\langle 0|\bar{s}\sigma_{\mu\nu}\gamma_5 c|D_{s1}^B(P,\eta)\rangle = f_T(\eta_\mu P_\nu - \eta_\nu P_\mu), \tag{44}$$

then contraction with $P^\nu/(m_c + m_s)$ gives the effective normalization

$$f_B^{\text{eff}} \simeq f_T\,\frac{m_{D_{s1}}}{m_c + m_s}, \tag{45}$$

up to the sign fixed by the precise convention for $\sigma_{\mu\nu}\gamma_5$ and by the transverse condition $\eta\cdot P = 0$.

Only after G_A and G_B have been computed with their own basis-current decay constants do we form the physical amplitudes:

$$G_{2460} = \sin\theta\, G_A + \cos\theta\, G_B, \tag{46}$$

$$G_{2536} = \cos\theta\, G_A - \sin\theta\, G_B. \tag{47}$$

Thus the sine–cosine factors belong to the physical-state mixing of amplitudes, not to an arbitrary identification $f_B = f_T$. In the numerical scripts, $f_B = f_T$ is kept only as a diagnostic. The converted $f_B = f_T m_{D_{s1}}/(m_c + m_s)$ option is the safer central normalization unless the input-table definition of f_T is changed.

10 Tensor-Current Hard Term: Triangle Reduction

The hard tensor-current numerator is more complicated than the axial one because the J_B current introduces an extra factor of P^α and therefore higher powers of the loop momentum after E1 projection. The script `scripts/step3_tensor_hard_triangle_reduction.wl` rewrites scalar products such as k^2, $k\cdot p$, and $k\cdot q$ in terms of inverse propagator denominators. For the heavy-line photon emission we define

$$d_{H1} = (k+q)^2 - m_Q^2, \tag{48}$$

$$d_{H2} = k^2 - m_Q^2, \tag{49}$$

$$d_{H3} = (k-p)^2 - m_s^2, \tag{50}$$

and similarly for the strange-line photon emission. Setting $d_{Hi} = 0$ isolates the genuine three-denominator triangle core. For the tensor current this gives

$$N_{B,H}^{\text{core}} = \frac{-8\mathrm{i}m_Q^2 + 4\mathrm{i}m_Q m_s}{m_Q + m_s} - \frac{4\mathrm{i}m_Q^2 p^2}{(m_Q + m_s)(p\cdot q)}, \tag{51}$$

$$N_{B,s}^{\text{core}} = \frac{-4\mathrm{i}m_Q m_s}{m_Q + m_s} - \frac{4\mathrm{i}m_s^2 p^2}{(m_Q + m_s)(p\cdot q)}. \tag{52}$$

The same reduction was also performed for the axial current in `scripts/step3_axial_hard_triangle_redu`
This is a useful calibration because the axial hard term has a known published spectral density.

10.1 Why denominator-cancellation terms are separated

Terms proportional to one or more inverse denominators cancel one or more propagators. After such a cancellation the loop integral contains at most two propagators. These reduced integrals depend on at most one of the two external virtualities, p^2 or $(p+q)^2$, and therefore do not generate the double-pole structure selected by the double Borel transform. They are recorded in the output files for auditability, but the double-pole sum rule is built from the three-denominator triangle core.

This observation narrows the remaining hard-tensor task: derive the double spectral density associated with the tensor-current triangle core and match it to the same E1 invariant amplitude used for the axial-current benchmark.

11 First Hard-Tensor Spectral Density Candidate

The next step is to translate the triangle core into a perturbative spectral density. We do this by calibrating against the known axial-current result. For one photon-emitting line, let m_i be the emitting-quark mass and m_j the spectator mass. The axial triangle core has the schematic form

$$N_i^{A,\mathrm{core}} = -4\mathrm{i}\left[a_i + \frac{b_i}{p\cdot q}\right]. \tag{53}$$

Comparing with the published axial spectral density fixes the replacement

$$-4\mathrm{i}\, a_i \longrightarrow 2a_i \log\frac{s-m_j^2+m_i^2-\lambda^{1/2}(s,m_j^2,m_i^2)}{s-m_j^2+m_i^2+\lambda^{1/2}(s,m_j^2,m_i^2)}, \tag{54}$$

$$-4\mathrm{i}\,\frac{b_i}{p\cdot q} \longrightarrow \frac{b_i}{m_i^2}\,\frac{m_j^2-m_i^2-s}{s^2}\,\lambda^{1/2}(s,m_j^2,m_i^2). \tag{55}$$

The overall factor is $-3/(8\pi^2)$, and the total axial density is assembled as the strange-emission contribution minus the heavy-emission contribution, with the corresponding charges. The script

`scripts/stage1_tensor_gb_hard_candidate.py`

first applies these rules back to the axial current and reproduces Colangelo's $\rho^P(s)$ with a maximum numerical difference 1.4×10^{-17} on the test grid.

For the tensor current the core contains $p^2/(p\cdot q)$:

$$N_{B,H}^{\mathrm{core}} = -4\mathrm{i}\left[\frac{2m_Q^2-m_Qm_s}{m_Q+m_s} + \frac{m_Q^2p^2}{(m_Q+m_s)(p\cdot q)}\right], \tag{56}$$

$$N_{B,s}^{\mathrm{core}} = -4\mathrm{i}\left[\frac{m_Qm_s}{m_Q+m_s} + \frac{m_s^2p^2}{(m_Q+m_s)(p\cdot q)}\right]. \tag{57}$$

The diagonal double-dispersion representation explains how the p^2 numerator is treated. For the double-pole part one has

$$\frac{p^2}{(s-p^2)(s-P^2)} = \frac{s}{(s-p^2)(s-P^2)} - \frac{1}{s-P^2}, \qquad P=p+q. \tag{58}$$

The second term has only a single pole and is removed by the double Borel projection in p^2 and P^2. Therefore, inside the double-pole sum rule, p^2 in the hard numerator can be replaced by the spectral variable s. The same argument gives $P^2\to s$ for any numerator factor of P^2. The script

`scripts/step4_double_pole_p2_reduction.py` records this identity as a short audit check.

Combining this hard candidate with the tensor-DA two-particle soft contribution gives, over the same M^2 and s_0 windows,

$$G_B^{\text{hard}} \simeq -0.831 \text{ to } -0.673 \text{ GeV}^{-1}, \quad G_B^{\text{Stage 1}} \simeq -0.510 \text{ to } -0.440 \text{ GeV}^{-1}, \quad (f_B = f_T), \tag{59}$$

$$G_B^{\text{hard}} \simeq -0.460 \text{ to } -0.373 \text{ GeV}^{-1}, \quad G_B^{\text{Stage 1}} \simeq -0.283 \text{ to } -0.244 \text{ GeV}^{-1}, \quad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s}\right). \tag{60}$$

At $\theta = 35.3^\circ$, the corresponding candidate mixed widths are

$$\Gamma[D_{s1}(2460) \to D_s\gamma] \simeq 62.5 \text{ to } 87.9 \text{ keV}, \quad \Gamma[D_{s1}(2536) \to D_s\gamma] \simeq 0.00 \text{ to } 0.30 \text{ keV}, \quad (f_B = f_T), \tag{61}$$

$$\Gamma[D_{s1}(2460) \to D_s\gamma] \simeq 30.9 \text{ to } 44.7 \text{ keV}, \quad \Gamma[D_{s1}(2536) \to D_s\gamma] \simeq 4.0 \text{ to } 8.2 \text{ keV}, \quad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s}\right). \tag{62}$$

These numbers should be read as a controlled Stage 1 candidate, not yet as the final result, because Stage 2 three-particle photon DAs remain. The $p^2 \to s$ step is justified for the double-pole part; any single-pole remnants belong outside the double-Borel sum rule.

12 Stage 2 Axial Three-Particle DA Check

Before deriving the tensor-current three-particle terms, we implemented the known axial-current correction as a Stage 2 calibration. The published axial sum rule contains the combination

$$\mathcal{F}_1 = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2), \tag{63}$$

integrated over the two standard regions

$$0 \le v \le 1 - u_0, \qquad 0 \le \alpha_g \le \frac{u_0}{1-v}, \tag{64}$$

$$1 - u_0 \le v \le 1, \qquad 0 \le \alpha_g \le \frac{1-u_0}{v}, \tag{65}$$

with

$$\alpha_q = u_0 - (1-v)\alpha_g, \qquad \alpha_{\bar{q}} = 1 - u_0 - v\alpha_g. \tag{66}$$

The script

`scripts/stage2_axial_g1_three_particle.py`

evaluates this integral using the three-particle photon DAs in `photon_da.py`. For $u_0 = 1/2$ it finds

$$\int \mathcal{F}_1 = +7.033 \times 10^{-2}, \tag{67}$$

$$\Delta G_A^{\mathcal{F}_1} \simeq +0.0012 \text{ to } +0.0020 \text{ GeV}^{-1}. \tag{68}$$

Thus the axial Stage 2 result over the same Borel and threshold window is

$$-0.398 \le G_A^{\text{Stage 2}} \le -0.312 \qquad \text{GeV}^{-1}, \tag{69}$$

corresponding to

$$20.5\,\text{keV} \le \Gamma_A^{\text{Stage 2}}[D_{s1}(2460) \to D_s\gamma] \le 33.3\,\text{keV}. \tag{70}$$

This small correction is a useful calibration, but it does not yet complete Stage 2 for the mixed physical amplitudes. The remaining Stage 2 task is to derive the tensor-current analog of the three-particle photon DA contribution from the background-gluon insertion and project it onto the same E1 invariant.

13 First Tensor-Current Stage 2 Trace Pass

For the tensor-current three-particle term we first separated the local $\sigma_{\alpha\beta}$ part of the heavy-quark background-gluon propagator from the explicitly $x_\alpha\gamma_\beta$ part. The script

`scripts/step5_soft_3p_trace_kernels.wl`

computes the E1-projected traces for the local piece against the $\mathcal{S}$, $\tilde{\mathcal{S}}$, $\mathcal{T}_1$, and $\mathcal{T}_2$ Lorentz structures. The nonzero local kernels are

$$K_A[\mathcal{S}] = 8\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{S}] = 8\,\frac{m_Q}{m_Q+m_s}, \tag{71}$$

$$K_A[\mathcal{T}_1] = -32\mathrm{i}\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{T}_1] = -32\mathrm{i}\,\frac{m_Q}{m_Q+m_s}, \tag{72}$$

$$K_A[\mathcal{T}_2] = +32\mathrm{i}\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{T}_2] = +32\mathrm{i}\,\frac{m_Q}{m_Q+m_s}. \tag{73}$$

The $\tilde{\mathcal{S}}$ kernel vanishes for this local part. After the three-particle light-cone Fourier integral and double Borel projection, $k\cdot q = p\cdot q$, so the local trace ratio is again

$$\frac{K_B}{K_A} = \frac{m_Q}{m_Q+m_s}. \tag{74}$$

This is the same ratio encountered in the two-particle tensor-DA family.

At this point it is useful to separate the published axial combination into the piece generated by the $\sigma_{\alpha\beta}$ part of the background-field propagator and the piece generated by the $x_\alpha G^{\alpha\beta}\gamma_\beta$ part:

$$\mathcal{F}_1^\sigma = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \tag{75}$$

$$\mathcal{F}_1^{x\gamma} = 2v\left(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2\right), \qquad \mathcal{F}_1 = \mathcal{F}_1^\sigma + \mathcal{F}_1^{x\gamma}. \tag{76}$$

The numerical split, evaluated in

`scripts/step10_stage2_f1_split.py`,

is

$$\int \mathcal{F}_1^\sigma = -1.5625\times 10^{-1}, \tag{77}$$

$$\int \mathcal{F}_1^{x\gamma} = +2.2658\times 10^{-1}, \tag{78}$$

$$\int \mathcal{F}_1 = +7.0330\times 10^{-2}. \tag{79}$$

Thus the small published axial Stage 2 correction is partly the result of a cancellation between two larger pieces. The simple trace ratio $m_Q/(m_Q+m_s)$ is established for the $\sigma_{\alpha\beta}$ part, so the controlled sigma-matched tensor-current correction is

$$\Delta G_B^{3p} \simeq -0.00366 \text{ to } -0.00210\ \mathrm{GeV}^{-1}, \qquad (f_B = f_T \text{ diagnostic}), \tag{80}$$

$$\Delta G_B^{3p} \simeq -0.00203 \text{ to } -0.00116\ \mathrm{GeV}^{-1}, \qquad \left(f_B = f_T\frac{m_{D_{s1}}}{m_c+m_s} \text{ central}\right). \tag{81}$$

The corresponding sigma-matched Stage 2 mixed-width estimate at $\theta = 35.3^\circ$ is

$$\Gamma[D_{s1}(2460)\to D_s\gamma] \simeq 62.9 \text{ to } 88.2\ \mathrm{keV}, \quad \Gamma[D_{s1}(2536)\to D_s\gamma] \simeq 0.00 \text{ to } 0.26\ \mathrm{keV}, \quad (f_B = f_T \text{ diagnostic}), \tag{82}$$

$$\Gamma[D_{s1}(2460)\to D_s\gamma] \simeq 30.9 \text{ to } 44.8\ \mathrm{keV}, \quad \Gamma[D_{s1}(2536)\to D_s\gamma] \simeq 3.8 \text{ to } 8.0\ \mathrm{keV}, \quad \Big(f_B = f_T\frac{m_{D_{s1}}}{m_c+m_s} \text{ centr} \tag{83}$$

If, instead, one multiplies the full axial $\mathcal{F}_1$ integral by the same trace ratio, one obtains only a diagnostic estimate:

$$\Delta G_B^{3p} \simeq +0.00094 \text{ to } +0.00165 \text{ GeV}^{-1}, \qquad (f_B = f_T \text{ diagnostic}), \tag{84}$$

$$\Delta G_B^{3p} \simeq +0.00052 \text{ to } +0.00091 \text{ GeV}^{-1}, \qquad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s} \text{ central}\right), \tag{85}$$

with central-normalization widths

$$\Gamma[D_{s1}(2460) \to D_s\gamma] \simeq 30.5 \text{ to } 44.5 \text{ keV}, \tag{86}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] \simeq 4.0 \text{ to } 8.1 \text{ keV}. \tag{87}$$

The difference between the sigma-matched and full-ratio diagnostic numbers is small at the current precision, but conceptually important: the $x_\alpha\gamma_\beta$ part must be matched separately before the tensor Stage 2 correction is final.

14 Stage 2 Tensor: Explicit x-Dependent Kernels

The next audit step keeps the coordinate vector x explicit. For the $\sigma_{\alpha\beta}$ part of S_Q^G, the raw E1 projections of the $\mathcal{T}_3$ and $\mathcal{T}_4$ structures are

$$K_A[\mathcal{T}_3] = +16\mathrm{i}\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{T}_3] = +16\mathrm{i}\,\frac{m_Q}{m_Q+m_s}, \tag{88}$$

$$K_A[\mathcal{T}_4] = -16\mathrm{i}\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{T}_4] = -16\mathrm{i}\,\frac{m_Q}{m_Q+m_s}. \tag{89}$$

Thus these two structures also obey the same ratio $K_B/K_A = m_Q/(m_Q+m_s)$ after $k\cdot q = p\cdot q$. This check is recorded in

`scripts/step6_soft_3p_x_dependent_kernels.wl`.

The genuinely new complication comes from the second term in the background propagator,

$$\frac{u\,x_\alpha}{m_Q^2-k^2}G^{\alpha\beta}(ux)\gamma_\beta. \tag{90}$$

The script

`scripts/step7_soft_3p_xgamma_kernels.wl`

computes the raw kernels with x kept explicit. Unlike the previous kernels, the tensor-to-axial ratio is not a constant at this stage. For example the $\mathcal{S}$ ratio after only $k\cdot q \to p\cdot q$ contains

$$\frac{(p\cdot q)\,(k\cdot x - p\cdot x) + (k\cdot p + p^2 + 2p\cdot q)(q\cdot x)}{2m_Q^2(q\cdot x)}, \tag{91}$$

and analogous expressions occur for $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$. Therefore the tensor Stage 2 three-particle result cannot yet be promoted from an estimate to a final term. The remaining calculation is to perform the Fourier/Borel replacement of the explicit $x_\alpha/(q\cdot x)$ factors, including the derivative action on the heavy-quark denominator, and then recombine the result with the photon DA structures.

As a diagnostic, we can impose only the light-cone momentum routing

$$k = p + aq, \qquad a = \alpha_{\bar q} + v\alpha_g, \qquad q^2 = 0, \tag{92}$$

without yet applying the derivative replacement. The script `scripts/step8_xgamma_saddle_reduction.py`

reduces the raw ratios $K_B/[(m_Q/(m_Q+m_s))K_A]$ to

$$R_{\mathcal{S}} = \frac{p^2+(a+1)p\cdot q}{m_Q^2}, \tag{93}$$

$$R_{\mathcal{T}_1} = \frac{p^2+(3a+2)p\cdot q}{3m_Q^2}, \tag{94}$$

$$R_{\mathcal{T}_2} = \frac{p^2+(2-a)p\cdot q}{m_Q^2}, \tag{95}$$

$$R_{\mathcal{T}_3} = \frac{p^2+2a\,p\cdot q}{m_Q^2}. \tag{96}$$

If one additionally uses the heavy-line on-shell relation $m_Q^2 = k^2 = p^2 + 2a\,p\cdot q$, only $R_{\mathcal{T}_3}$ becomes exactly one. The other ratios remain nontrivial. This confirms that the $x_\alpha\gamma_\beta$ term must be treated with the full Fourier/Borel derivative machinery. It is not controlled by the simple trace-ratio argument used for the $\sigma_{\alpha\beta}$ part.

15 Stage 2 Tensor: Derivative Reduction of $x_\alpha\gamma_\beta$

The next step applies the Fourier derivative rules to the raw $x_\alpha\gamma_\beta$ kernels. With phase

$$\exp[\mathrm{i}(p+aq-k)\cdot x], \tag{97}$$

integration over x gives, up to a common phase convention,

$$p\cdot x \to -\mathrm{i}\,p\cdot\partial_k, \tag{98}$$

$$q\cdot x \to -\mathrm{i}\,q\cdot\partial_k, \tag{99}$$

$$k\cdot x\,F(k) \to -\mathrm{i}\,\partial_{k_\mu}[k_\mu F(k)] = -\mathrm{i}\,[4F(k) + k\cdot\partial_k F(k)]\,. \tag{100}$$

The script

`scripts/step9_xgamma_derivative_reduction.wl`

applies these rules to the scalar kernels multiplying $1/(m_Q^2-k^2)$. After the light-cone routing $k = p + aq$, the reduced axial kernels are proportional to

$$A_{\mathcal{S}}^{x\gamma} = \frac{-8m_Q}{[-m_Q^2+p^2+2a\,p\cdot q]^2}, \tag{101}$$

$$A_{\mathcal{T}_1}^{x\gamma} = \frac{48\mathrm{i}m_Q}{[-m_Q^2+p^2+2a\,p\cdot q]^2}, \tag{102}$$

$$A_{\mathcal{T}_2}^{x\gamma} = \frac{16\mathrm{i}m_Q}{[-m_Q^2+p^2+2a\,p\cdot q]^2}, \tag{103}$$

$$A_{\mathcal{T}_3}^{x\gamma} = \frac{-16\mathrm{i}m_Q}{[-m_Q^2+p^2+2a\,p\cdot q]^2}. \tag{104}$$

The corresponding tensor-current kernels are not obtained by a universal $m_Q/(m_Q+m_s)$ multiplication. The ratios after derivative reduction are

$$R^{\partial}_{\mathcal{S}} = 1 + \frac{(1-a)p\cdot q}{m_Q^2}, \tag{105}$$

$$R^{\partial}_{\mathcal{T}_1} = \frac{2m_Q^2 + p^2 + (2-a)p\cdot q}{3m_Q^2}, \tag{106}$$

$$R^{\partial}_{\mathcal{T}_2} = -2 + \frac{3p^2 + (2+3a)p\cdot q}{m_Q^2}, \tag{107}$$

$$R^{\partial}_{\mathcal{T}_3} = \frac{2m_Q^2 - p^2 - 2a\,p\cdot q}{m_Q^2}. \tag{108}$$

All these reduced kernels carry the squared heavy-quark denominator. For the double-pole residue we drop numerator terms proportional to $D = -m_Q^2 + p^2 + 2a\,p\cdot q$, equivalently using $m_Q^2 = p^2 + 2a\,p\cdot q$ in the residue. In the three-particle integration domains,

$$a = \alpha_{\bar q} + v\alpha_g = 1 - u_0, \tag{109}$$

so for the symmetric Borel point $u_0 = 1/2$ this residue is evaluated at $a = 1/2$.

The script

`scripts/step11_tensor_xgamma_ratio_integral.py`

uses these residue ratios to match the $\mathcal{S}$, $\mathcal{T}_2$, and $\mathcal{T}_3$ pieces of the tensor-current $x_\alpha\gamma_\beta$ contribution to the known axial normalization

$$\mathcal{F}_1^{x\gamma} = 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \tag{110}$$

With the heavy-line residue condition and physical $p\cdot q$ in the E1 invariant, the matched $\mathcal{S}, \mathcal{T}_2, \mathcal{T}_3$ integral is

$$\int \mathcal{F}^{x\gamma}_{1,B}[\mathcal{S}, \mathcal{T}_2, \mathcal{T}_3] = +2.5818 \times 10^{-1}. \tag{111}$$

The tensor-current derivative reduction also produces a $\mathcal{T}_4$ residue even though the axial E1 kernel has $A^{x\gamma}_{\mathcal{T}_4} = 0$. This is not an obstacle: its normalization can be fixed directly relative to the axial $\mathcal{T}_2$ unit. On the double-pole residue,

$$\frac{B^{x\gamma}_{\mathcal{T}_4}}{[m_Q/(m_Q+m_s)]\,A^{x\gamma}_{\mathcal{T}_2}} = -\frac{(1-a)p\cdot q}{m_Q^2}. \tag{112}$$

This gives the tensor-only contribution

$$\int \mathcal{F}^{x\gamma}_{1,B}[\mathcal{T}_4] = -4.2136 \times 10^{-2}, \tag{113}$$

and therefore

$$\int \mathcal{F}^{x\gamma}_{1,B} = +2.1605 \times 10^{-1}. \tag{114}$$

Together with the sigma-like part this gives the full matched tensor three-particle integral

$$\int \left(\mathcal{F}^{\sigma}_{1,B} + \mathcal{F}^{x\gamma}_{1,B}\right) = +5.9796 \times 10^{-2}. \tag{115}$$

For comparison, the axial total is $\int \mathcal{F}_1 = +7.0330 \times 10^{-2}$. A diagnostic calculation that instead inserts the physical hadron value $p^2 = m_{D_s}^2$ directly into the OPE residue ratios gives a much larger integral, $+1.5072$, so we do not use it as the working tensor estimate.

Numerically, for the central decay-constant normalization $f_B = f_T m_{D_{s1}}/(m_c + m_s)$, the full matched Stage 2 tensor-current result gives

$$\Delta G_B^{3p} = +0.00044 \text{ to } +0.00078 \text{ GeV}^{-1}, \tag{116}$$

$$G_B^{\text{Stage 2}} = -0.2821 \text{ to } -0.2436 \text{ GeV}^{-1}. \tag{117}$$

At $\theta = 35.3°$ this corresponds to

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 30.6 \text{ to } 44.5 \text{ keV}, \tag{118}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 4.0 \text{ to } 8.1 \text{ keV}. \tag{119}$$

The final matched result is close to the older full-ratio diagnostic, but the derivation is now explicit about which pieces come from the $\sigma_{\alpha\beta}$ part, which come from $x_\alpha G^{\alpha\beta}\gamma_\beta$, and how the tensor-only $\mathcal{T}_4$ term is normalized.

16 Final Stage 2 Uncertainty Scan

The script

`scripts/final_stage2_uncertainty_scan.py`

performs a first independent-input uncertainty scan around the completed matched Stage 2 baseline. Each random point samples the Borel parameter $M^2 \in [3, 6]$ GeV^2, the continuum threshold $\sqrt{s_0} \in [2.50, 2.60]$ GeV, and the numerical inputs listed in `inputs_table.csv`. The tensor-current decay constant is converted as

$$f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s}. \tag{120}$$

The scan is not yet a correlated global error analysis; it is a transparent first error budget for the present sum rule.

For fixed $\theta = 35.3°$, using 400 random points, the scan gives

$$G[D_{s1}(2460) \to D_s\gamma] = -0.427^{+0.065}_{-0.063} \text{ GeV}^{-1}, \tag{121}$$

$$G[D_{s1}(2536) \to D_s\gamma] = -0.140^{+0.045}_{-0.053} \text{ GeV}^{-1}, \tag{122}$$

where the uncertainty indicates the 16–84 percentile interval. The corresponding width estimates are

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 38.3^{+12.3}_{-10.7} \text{ keV}, \tag{123}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 6.1^{+5.5}_{-3.3} \text{ keV}. \tag{124}$$

If instead the mixing angle is sampled uniformly over $25° \le \theta \le 45°$, the same scan gives

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 34.6^{+12.8}_{-8.7} \text{ keV}, \tag{125}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 4.9^{+7.1}_{-3.6} \text{ keV}. \tag{126}$$

The full scan table is written to

`outputs/final_stage2_uncertainty_scan.csv`.

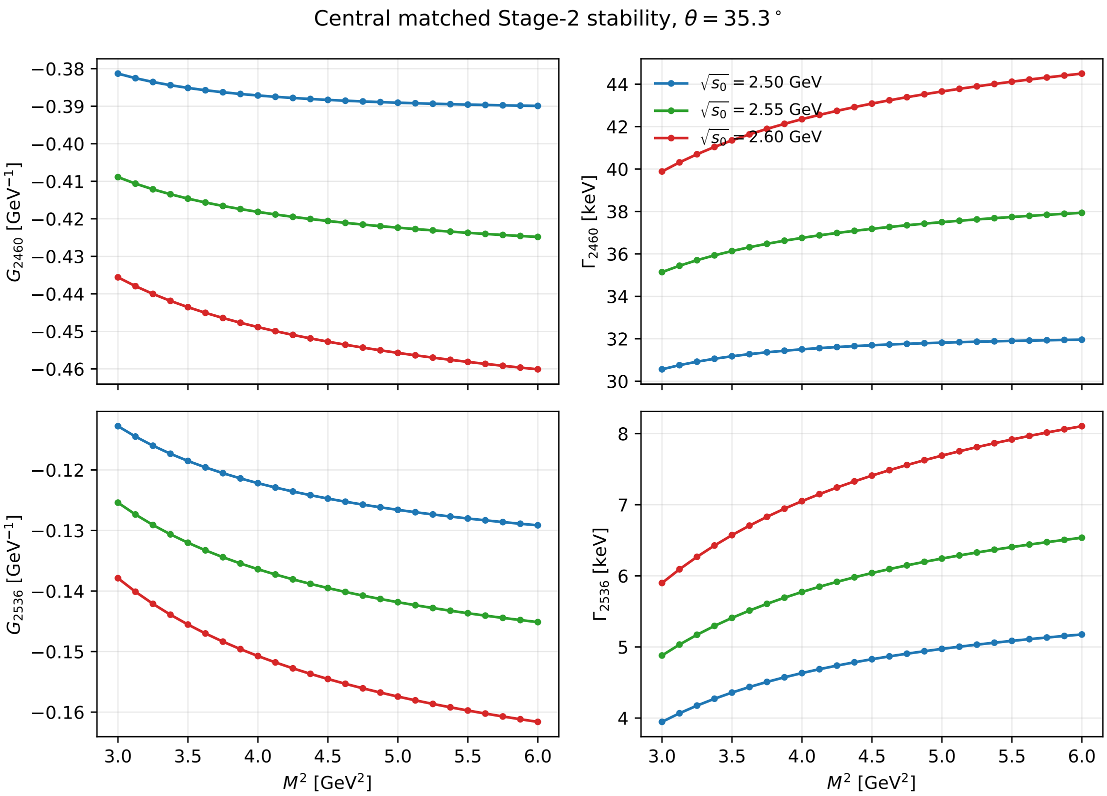

Figure 1: Central matched Stage 2 stability for $D_{s1}(2460) \to D_s\gamma$ and $D_{s1}(2536) \to D_s\gamma$ at $\theta = 35.3°$. The three curves correspond to $\sqrt{s_0} = 2.50, 2.55, 2.60$ GeV.

17 Central Borel Stability

For the central input set and $\theta = 35.3°$, the script

`scripts/stage2_stability_plots.py`

produces the Borel-stability curves shown in Fig. 1. The curves are smooth over $3 \le M^2 \le 6$ GeV^2. At fixed $\sqrt{s_0}$, the dependence on M^2 is mild; changing $\sqrt{s_0}$ from 2.50 to 2.60 GeV shifts the normalization more visibly and should therefore remain part of the quoted systematic uncertainty.

Across the full displayed window the central ranges are

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 30.6 \text{ to } 44.5 \text{ keV}, \tag{127}$$
$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 3.95 \text{ to } 8.10 \text{ keV}. \tag{128}$$

The corresponding amplitude spans are smaller:

$$G_{2460} = -0.460 \text{ to } -0.381 \text{ GeV}^{-1}, \tag{129}$$
$$G_{2536} = -0.162 \text{ to } -0.113 \text{ GeV}^{-1}. \tag{130}$$

This is the expected pattern: the widths amplify the residual sum-rule variation because $\Gamma \propto G^2|\vec{q}|^3$.

18 Publication Figures and Phenomenology

The remaining publication-oriented figures are generated by

`scripts/publication_plots.py`.

Figure 2 shows the continuum-threshold dependence at fixed representative values of M^2. It complements Fig. 1: the curves are monotonic in $\sqrt{s_0}$, so the continuum threshold should be quoted as an important systematic.

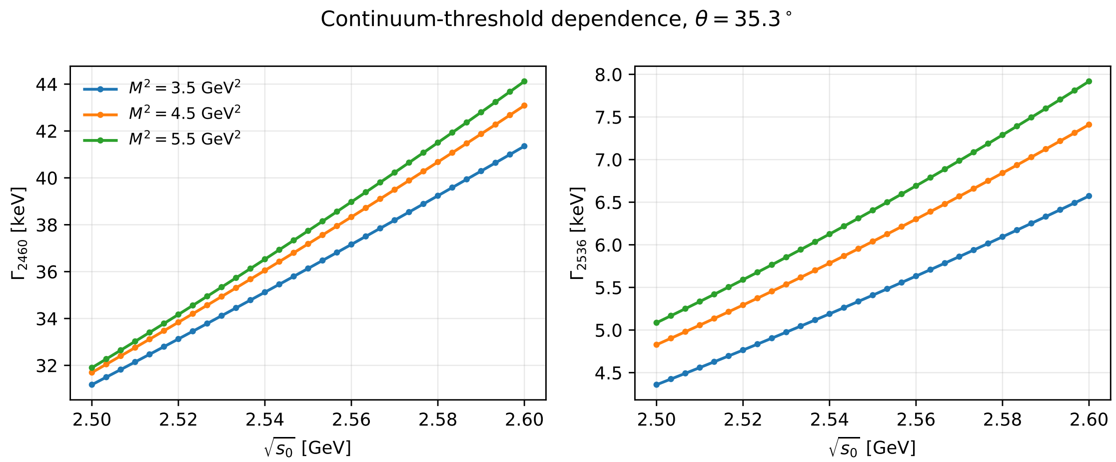

Figure 2: Continuum-threshold dependence of the matched Stage 2 widths for $\theta = 35.3°$ at three representative Borel points.

Figure 3 shows the mixing-angle dependence. The $D_{s1}(2460)$ width remains at the tens-of-keV level over a broad range of θ, while the $D_{s1}(2536)$ width is strongly suppressed near the large-angle region because its amplitude involves a cancellation between the axial and tensor-current components.

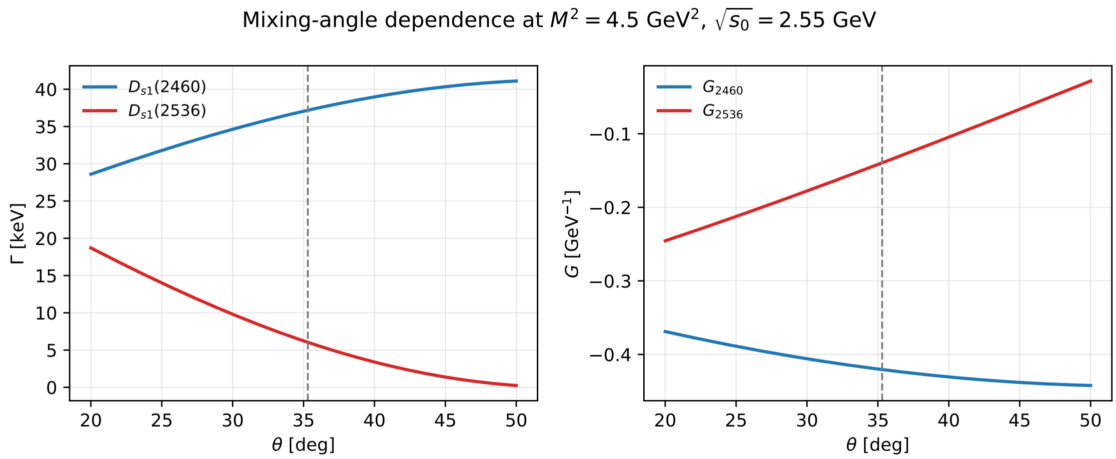

Figure 3: Mixing-angle dependence at $M^2 = 4.5$ GeV^2 and $\sqrt{s_0} = 2.55$ GeV. The dashed line marks $\theta = 35.3°$.

The cancellation pattern is displayed more explicitly in Fig. 4. At the central point $M^2 = 4.5$ GeV^2, $\sqrt{s_0} = 2.55$ GeV, the $D_{s1}(2536)$ amplitude receives opposite-sign contributions from the axial Stage 1 term and the hard tensor-current term. The three-particle DA corrections are numerically small in the final physical amplitudes, but they are included in the quoted Stage 2 result.

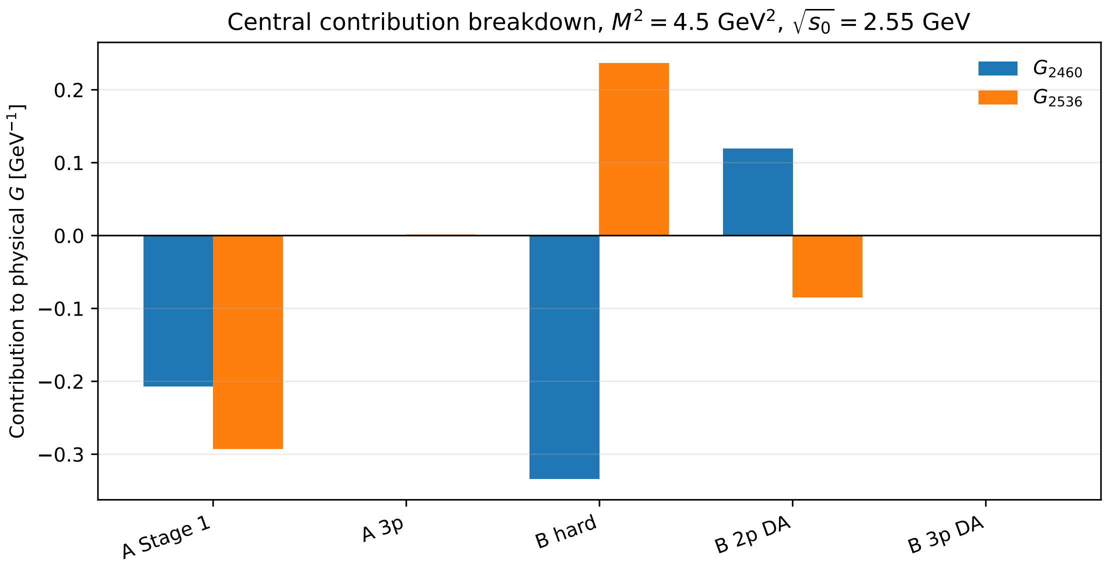

Figure 4: Contribution breakdown for the physical couplings at the central Borel and threshold point. The labels “A” and “B” refer to the axial-current and tensor-current components before physical mixing.

Finally, Fig. 5 compares the present result for $D_{s1}(2460) \to D_s\gamma$ with representative estimates collected by Colangelo, De Fazio and Ozpineci. The present result is higher than the original axial-current LCSR interval, but it remains the same order of magnitude. It should not be advertised as explaining a non-observation by making the $D_{s1}(2460) \to D_s\gamma$ width tiny. Instead, the robust phenomenological statement from this calculation is that the $D_{s1}(2536) \to D_s\gamma$ channel is much more cancellation-sensitive and can be substantially smaller.

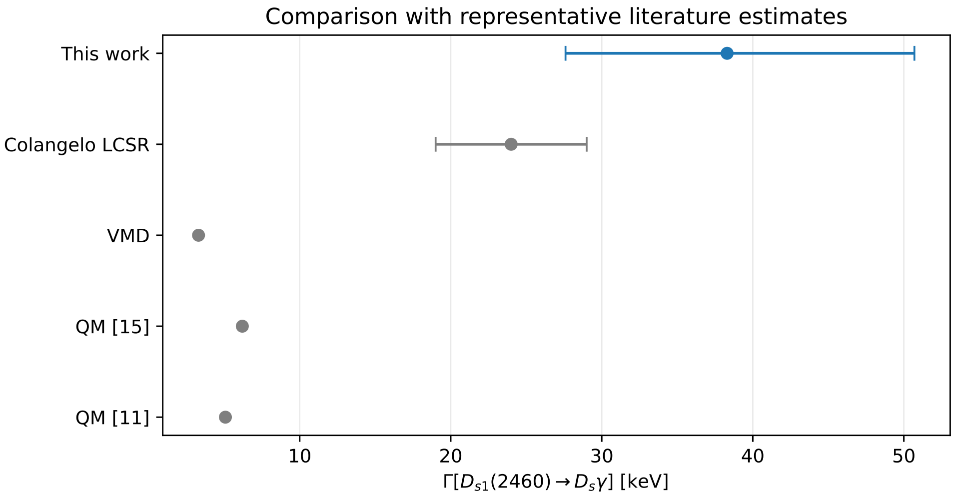

Figure 5: Comparison of the present $D_{s1}(2460) \to D_s\gamma$ width with representative literature estimates. The Colangelo LCSR, VMD and quark-model values are from the comparison table in Colangelo, De Fazio and Ozpineci (hep-ph/0505195).